

DATA CLEANING & STRUCTURAL VALIDATION

COMPLETE TASK SUBMISSION

Dataset: Customer Sales Dataset

Task Requirement	Completed Work
Download / use a sample dataset	Customer Sales dataset prepared for this task
Inspect missing values, duplicates and data types	Completed using Python Pandas
Handle missing values	Mode / median / median-date imputation
Standardize headers, dates and categories	Completed
Export clean data	Cleaned dataset included in this PDF appendix

This single PDF contains the written task answer, methodology, validation results, Python code, and the final cleaned dataset.

1. Objective

The objective of this task is to transform raw, unstructured customer sales data into a clean and reliable format for analysis. The dataset was checked for duplicate records, missing values, inconsistent text, mixed date formats, incorrect data types, and invalid numerical values. Python Pandas and NumPy were used to perform the cleaning and structural validation.

2. Dataset Used

The Customer Sales dataset contains 100 final records and 11 final columns. It contains customer, location, product, order, payment and sales information. The raw version used for this task intentionally contained realistic data-quality issues so that each cleaning requirement could be demonstrated.

Column	Purpose
order_id	Unique order identifier
customer_name	Name of the customer
city	Customer city
product	Purchased product
category	Product category
order_date	Date on which the order was placed
quantity	Number of units purchased
unit_price	Price of one unit
total_amount	Calculated as quantity × unit price
payment_method	Payment method used by the customer
order_status	Order status such as Completed, Pending or Cancelled

3. Data Problems Found and Solutions

Problem Found	Solution Applied	Reason
Duplicate records	Removed exact duplicate rows	Prevents the same order from being counted more than once
Missing categorical values	Filled with mode	Keeps the records while using the most common valid category
Missing numeric values	Filled with median	Median is less affected by extreme values
Missing date	Filled with median date	Keeps the record usable for date-based analysis
Extra spaces / inconsistent capitalization	Trimmed spaces and standardized capitalization	Creates consistent categories and names
Mixed date formats	Converted to datetime	Provides one consistent date structure
Numbers stored as text	Converted with <code>pd.to_numeric()</code>	Ensures numerical calculations work correctly
Invalid negative price	Changed to missing, then imputed	Negative product price is not valid for this dataset

4. Step-by-Step Data Cleaning Process

Step 1 — Load the dataset — The raw CSV file was loaded into a Pandas DataFrame.

Step 2 — Inspect the dataset — The shape, first rows, data types, missing values and duplicate rows were checked.

- Step 3 — Standardize column headers** — Column names were converted to lowercase and spaces were replaced with underscores.
- Step 4 — Clean text values** — Leading/trailing spaces were removed and names/categories were standardized.
- Step 5 — Convert dates** — Different date formats were converted into a consistent datetime representation.
- Step 6 — Convert numeric values** — Quantity and Unit Price were converted to numeric data types.
- Step 7 — Handle invalid values** — Impossible values such as a negative price were treated as missing.
- Step 8 — Handle missing values** — Categorical values were filled using mode and numeric values using median.
- Step 9 — Remove duplicates** — Exact duplicate records were removed.
- Step 10 — Create Total Amount** — Total Amount was calculated using $\text{Quantity} \times \text{Unit Price}$.
- Step 11 — Validate** — Missing values, duplicates, duplicate IDs and invalid numeric values were checked again.
- Step 12 — Export** — The final cleaned dataset was exported as a CSV.

5. Python Code Used

```
import pandas as pd
import numpy as np

# Load data
df = pd.read_csv("customer_sales_messy.csv")

# Inspect
print(df.shape)
print(df.isnull().sum())
print("Duplicates:", df.duplicated().sum())
print(df.dtypes)

# Standardize column names
df.columns = (
df.columns.str.strip()
.str.lower()
.str.replace(" ", "_", regex=False)
)

# Clean text
text_cols = df.select_dtypes(include="object").columns
for col in text_cols:
df[col] = df[col].astype("string").str.strip()

df["customer_name"] = (
df["customer_name"]
.str.replace(r"\s+", " ", regex=True)
.str.title()
)

for col in ["city", "product", "category", "payment_method", "order_status"]:
df[col] = df[col].str.title()

# Convert dates and numeric columns
df["order_date"] = pd.to_datetime(
df["order_date"], errors="coerce", dayfirst=True
)
df["quantity"] = pd.to_numeric(df["quantity"], errors="coerce")
df["unit_price"] = pd.to_numeric(df["unit_price"], errors="coerce")

# Treat impossible values as missing
df.loc[df["quantity"] <= 0, "quantity"] = np.nan
df.loc[df["unit_price"] <= 0, "unit_price"] = np.nan

# Fill categorical missing values with mode
for col in ["city", "customer_name", "payment_method"]:
df[col] = df[col].fillna(df[col].mode()[0])

# Fill numerical missing values with median
df["quantity"] = df["quantity"].fillna(df["quantity"].median()).round().astype(int)
df["unit_price"] = df["unit_price"].fillna(df["unit_price"].median())

# Fill missing date with median date
valid_dates = df["order_date"].dropna().sort_values()
median_date = valid_dates.iloc[len(valid_dates)//2]
df["order_date"] = df["order_date"].fillna(median_date)

# Remove duplicates
df = df.drop_duplicates().reset_index(drop=True)

# Calculate total amount
df["total_amount"] = df["quantity"] * df["unit_price"]

# Final validation
print("Missing values:", df.isnull().sum().sum())
print("Duplicate rows:", df.duplicated().sum())
print("Duplicate Order IDs:", df["order_id"].duplicated().sum())

# Export
df.to_csv("customer_sales_cleaned.csv", index=False)
```

6. Final Validation

Check	Result	Status
Raw records	103	Completed
Final cleaned records	100	PASS
Duplicate rows after cleaning	0	PASS
Missing values after cleaning	0	PASS
Duplicate Order IDs	0	PASS
Invalid unit prices	0	PASS
Invalid quantities	0	PASS

7. Conclusion

The raw customer sales dataset was successfully cleaned and structurally validated. Duplicate records were removed, missing values were handled using documented rules, text and date formats were standardized, numeric fields were validated, and a Total Amount field was created. The final dataset is consistent and ready for further analysis. This PDF contains the complete written answer and a copy of the final cleaned dataset below.

APPENDIX A — FINAL CLEANED DATASET

The following table is the final cleaned dataset. It is included in this single PDF so that the submission contains both the written task answer and the cleaned data.

order_id	customer_name	city	product	category	order_date	quantity	unit_price	total_amount	payment_method	order_status
ORD0001	Rahul Nair	Bhopal	Laptop	Electronics	2026-08-17	3	55000	165000	Net Banking	Completed
ORD0002	Sneha Rao	Jaipur	Laptop	Electronics	2026-02-20	1	55000	55000	Net Banking	Pending
ORD0003	Ananya Singh	Ahmedabad	Smart Watch	Wearables	2026-02-25	3	5000	15000	Debit Card	Completed
ORD0004	Priya Patel	Jaipur	Laptop	Electronics	2026-02-01	3	62000	186000	Cash	Completed
ORD0005	Karan Jain	Pune	Monitor	Electronics	2026-07-17	3	9000	27000	Debit Card	Completed
ORD0006	Meera Das	Indore	Monitor	Electronics	2026-04-01	1	13000	13000	Credit Card	Cancelled
ORD0007	Vikram Joshi	Bhopal	Smart Watch	Wearables	2026-08-25	2	5000	10000	Net Banking	Completed
ORD0008	Rahul Nair	Bengaluru	Monitor	Electronics	2026-08-22	1	9000	9000	Net Banking	Cancelled
ORD0009	Neha Verma	Mumbai	Keyboard	Accessories	2026-07-10	4	800	3200	Net Banking	Cancelled
ORD0010	Vikram Joshi	Jaipur	Smart Watch	Wearables	2026-08-09	4	5000	20000	Upi	Completed
ORD0011	Rahul Nair	Delhi	Monitor	Electronics	2026-03-06	4	18000	72000	Net Banking	Completed
ORD0012	Rahul Nair	Bhopal	Keyboard	Accessories	2026-01-01	3	800	2400	Debit Card	Cancelled
ORD0013	Aditya Kapoor	Indore	Smart Watch	Wearables	2026-04-06	1	8000	8000	Credit Card	Cancelled
ORD0014	Aditya Kapoor	Indore	Smart Watch	Wearables	2026-03-03	3	5000	15000	Net Banking	Completed
ORD0015	Priya Patel	Ahmedabad	Monitor	Electronics	2026-03-09	2	9000	18000	Net Banking	Cancelled
ORD0016	Arjun Gupta	Mumbai	Smart Watch	Wearables	2026-08-19	2	8000	16000	Net Banking	Pending
ORD0017	Priya Patel	Mumbai	Smartphone	Electronics	2026-01-02	1	18000	18000	Upi	Cancelled
ORD0018	Rahul Nair	Indore	Smartphone	Electronics	2026-05-05	1	25000	25000	Credit Card	Cancelled
ORD0019	Rohan Mehta	Indore	Smartphone	Electronics	2026-04-19	4	18000	72000	Cash	Pending
ORD0020	Meera Das	Indore	Keyboard	Accessories	2026-02-18	1	800	800	Net Banking	Pending
ORD0021	Rohan Mehta	Jaipur	Smartphone	Electronics	2026-05-19	3	18000	54000	Net Banking	Completed
ORD0022	Priya Patel	Mumbai	Smartphone	Electronics	2026-01-01	4	25000	100000	Cash	Pending
ORD0023	Kabir Malhotra	Ahmedabad	Headphones	Accessories	2026-02-18	4	4000	16000	Debit Card	Completed
ORD0024	Aarav Sharma	Indore	Keyboard	Accessories	2026-01-15	3	2000	6000	Net Banking	Completed
ORD0025	Meera Das	Delhi	Laptop	Electronics	2026-06-02	1	70000	70000	Upi	Cancelled
ORD0026	Priya Patel	Jaipur	Keyboard	Accessories	2026-04-12	3	800	2400	Credit Card	Cancelled

Comprehensive Order Management System - Q3 2026										
order_id	customer_name	city	product	category	order_date	quantity	unit_price	total_amount	payment_method	order_status
ORD0027	Rahul Nair	Pune	Tablet	Electronics	2026-05-18	3	30000	90000	Credit Card	Cancelled
ORD0028	Vikram Joshi	Delhi	Headphones	Accessories	2026-03-19	1	2500	2500	Net Banking	Cancelled
ORD0029	Priya Patel	Indore	Keyboard	Accessories	2026-05-22	3	800	2400	Credit Card	Pending
ORD0030	Vikram Joshi	Mumbai	Keyboard	Accessories	2026-08-20	3	1200	3600	Upi	Completed
ORD0031	Rahul Nair	Jaipur	Monitor	Electronics	2026-07-09	3	9000	27000	Cash	Cancelled
ORD0032	Rahul Nair	Jaipur	Smart Watch	Wearables	2026-05-30	1	8000	8000	Net Banking	Completed
ORD0033	Arjun Gupta	Delhi	Laptop	Electronics	2026-01-27	3	62000	186000	Debit Card	Cancelled
ORD0034	Kabir Malhotra	Jaipur	Smart Watch	Wearables	2026-08-25	2	3000	6000	Upi	Pending
ORD0035	Aditya Kapoor	Mumbai	Headphones	Accessories	2026-07-29	2	2500	5000	Cash	Pending
ORD0036	Vikram Joshi	Mumbai	Smartphone	Electronics	2026-03-31	1	18000	18000	Net Banking	Pending
ORD0037	Rahul Nair	Bengaluru	Laptop	Electronics	2026-04-22	1	62000	62000	Debit Card	Cancelled
ORD0038	Aditya Kapoor	Bengaluru	Tablet	Electronics	2026-01-01	1	13000	13000	Net Banking	Cancelled
ORD0039	Rahul Nair	Mumbai	Headphones	Accessories	2026-07-04	4	2500	10000	Debit Card	Cancelled
ORD0040	Vikram Joshi	Jaipur	Headphones	Accessories	2026-08-29	4	4000	16000	Cash	Cancelled
ORD0041	Pooja Yadav	Delhi	Smart Watch	Wearables	2026-07-21	3	5000	15000	Net Banking	Cancelled
ORD0042	Rahul Nair	Bengaluru	Tablet	Electronics	2026-05-19	4	30000	120000	Upi	Pending
ORD0043	Karan Jain	Bengaluru	Laptop	Electronics	2026-03-04	2	70000	140000	Cash	Cancelled
ORD0044	Meera Das	Bhopal	Tablet	Electronics	2026-04-13	4	30000	120000	Credit Card	Completed
ORD0045	Rahul Nair	Indore	Monitor	Electronics	2026-05-23	1	9000	9000	Cash	Completed
ORD0046	Ishita Shah	Delhi	Monitor	Electronics	2026-07-28	4	18000	72000	Net Banking	Pending
ORD0047	Meera Das	Ahmedabad	Smartphone	Electronics	2026-06-10	4	25000	100000	Credit Card	Pending
ORD0048	Ishita Shah	Bhopal	Keyboard	Accessories	2026-02-08	3	800	2400	Credit Card	Pending
ORD0049	Pooja Yadav	Delhi	Keyboard	Accessories	2026-01-16	2	1200	2400	Credit Card	Pending
ORD0050	Arjun Gupta	Indore	Monitor	Electronics	2026-07-11	4	18000	72000	Credit Card	Cancelled
ORD0051	Kabir Malhotra	Mumbai	Tablet	Electronics	2026-06-23	2	16000	32000	Cash	Cancelled
ORD0052	Rohan Mehta	Ahmedabad	Smartphone	Electronics	2026-04-20	1	32000	32000	Credit Card	Pending
ORD0053	Rohan Mehta	Indore	Headphones	Accessories	2026-07-12	4	2500	10000	Cash	Pending
ORD0054	Meera Das	Bhopal	Tablet	Electronics	2026-07-19	3	22000	66000	Upi	Completed
ORD0055	Ananya Singh	Mumbai	Monitor	Electronics	2026-05-25	1	9000	9000	Credit Card	Pending

Comprehensive Order Management System - Q3 2026										
order_id	customer_name	city	product	category	order_date	quantity	unit_price	total_amount	payment_method	order_status
ORD0056	Pooja Yadav	Pune	Monitor	Electronics	2026-03-21	3	9000	27000	Upi	Cancelled
ORD0057	Aarav Sharma	Pune	Smart Watch	Wearables	2026-03-04	4	8000	32000	Upi	Cancelled
ORD0058	Aditya Kapoor	Pune	Monitor	Electronics	2026-01-18	1	13000	13000	Net Banking	Cancelled
ORD0059	Neha Verma	Indore	Monitor	Electronics	2026-02-09	4	18000	72000	Cash	Completed
ORD0060	Pooja Yadav	Pune	Smart Watch	Wearables	2026-08-01	4	5000	20000	Upi	Pending
ORD0061	Kabir Malhotra	Ahmedabad	Smartphone	Electronics	2026-08-06	4	32000	128000	Debit Card	Completed
ORD0062	Ishita Shah	Mumbai	Headphones	Accessories	2026-05-13	3	4000	12000	Credit Card	Completed
ORD0063	Ananya Singh	Jaipur	Tablet	Electronics	2026-01-24	2	30000	60000	Debit Card	Completed
ORD0064	Arjun Gupta	Mumbai	Headphones	Accessories	2026-03-26	3	4000	12000	Debit Card	Cancelled
ORD0065	Ishita Shah	Pune	Headphones	Accessories	2026-05-18	3	4000	12000	Credit Card	Completed
ORD0066	Ananya Singh	Ahmedabad	Headphones	Accessories	2026-01-04	3	2500	7500	Net Banking	Completed
ORD0067	Vikram Joshi	Indore	Laptop	Electronics	2026-05-19	3	70000	210000	Cash	Pending
ORD0068	Ishita Shah	Bengaluru	Smartphone	Electronics	2026-05-28	1	25000	25000	Upi	Completed
ORD0069	Rohan Mehta	Pune	Laptop	Electronics	2026-07-18	2	55000	110000	Net Banking	Pending
ORD0070	Rohan Mehta	Ahmedabad	Headphones	Accessories	2026-07-15	4	4000	16000	Credit Card	Cancelled
ORD0071	Ananya Singh	Pune	Keyboard	Accessories	2026-04-15	1	800	800	Cash	Cancelled
ORD0072	Sneha Rao	Ahmedabad	Headphones	Accessories	2026-06-25	1	4000	4000	Credit Card	Pending
ORD0073	Aditya Kapoor	Mumbai	Tablet	Electronics	2026-01-16	3	16000	48000	Credit Card	Pending
ORD0074	Sneha Rao	Pune	Tablet	Electronics	2026-05-19	1	30000	30000	Cash	Pending
ORD0075	Priya Patel	Indore	Tablet	Electronics	2026-05-31	3	16000	48000	Upi	Cancelled
ORD0076	Meera Das	Pune	Smart Watch	Wearables	2026-05-30	1	5000	5000	Cash	Cancelled
ORD0077	Meera Das	Ahmedabad	Laptop	Electronics	2026-03-26	4	62000	248000	Cash	Pending
ORD0078	Ishita Shah	Pune	Keyboard	Accessories	2026-01-30	4	1200	4800	Cash	Cancelled
ORD0079	Meera Das	Indore	Keyboard	Accessories	2026-06-10	4	2000	8000	Cash	Completed
ORD0080	Ananya Singh	Pune	Smart Watch	Wearables	2026-06-05	2	3000	6000	Credit Card	Cancelled
ORD0081	Rohan Mehta	Pune	Smart Watch	Wearables	2026-06-15	1	3000	3000	Credit Card	Pending
ORD0082	Kabir Malhotra	Pune	Smart Watch	Wearables	2026-04-09	3	5000	15000	Credit Card	Cancelled
ORD0083	Karan Jain	Delhi	Monitor	Electronics	2026-07-05	1	13000	13000	Debit Card	Cancelled
ORD0084	Sneha Rao	Ahmedabad	Monitor	Electronics	2026-04-07	2	9000	18000	Cash	Pending

order_id	customer_name	city	product	category	order_date	quantity	unit_price	total_amount	payment_method	order_status
ORD0085	Meera Das	Mumbai	Keyboard	Accessories	2026-05-02	2	1200	2400	Cash	Pending
ORD0086	Rahul Nair	Delhi	Tablet	Electronics	2026-05-15	1	22000	22000	Cash	Completed
ORD0087	Pooja Yadav	Delhi	Tablet	Electronics	2026-06-16	2	22000	44000	Upi	Pending
ORD0088	Meera Das	Jaipur	Headphones	Accessories	2026-08-24	4	4000	16000	Debit Card	Completed
ORD0089	Pooja Yadav	Bhopal	Tablet	Electronics	2026-05-19	1	22000	22000	Upi	Completed
ORD0090	Neha Verma	Indore	Headphones	Accessories	2026-06-24	3	1500	4500	Credit Card	Completed
ORD0091	Rohan Mehta	Bhopal	Smart Watch	Wearables	2026-04-29	4	8000	32000	Debit Card	Pending
ORD0092	Vikram Joshi	Indore	Smartphone	Electronics	2026-05-31	4	13000	52000	Debit Card	Cancelled
ORD0093	Arjun Gupta	Pune	Tablet	Electronics	2026-04-02	3	16000	48000	Net Banking	Pending
ORD0094	Pooja Yadav	Pune	Laptop	Electronics	2026-07-10	4	62000	248000	Cash	Pending
ORD0095	Aditya Kapoor	Mumbai	Smart Watch	Wearables	2026-07-16	3	5000	15000	Credit Card	Cancelled
ORD0096	Priya Patel	Indore	Headphones	Accessories	2026-03-25	4	2500	10000	Cash	Completed
ORD0097	Ananya Singh	Delhi	Monitor	Electronics	2026-07-11	3	13000	39000	Debit Card	Completed
ORD0098	Ishita Shah	Bhopal	Smartphone	Electronics	2026-07-22	2	32000	64000	Cash	Completed
ORD0099	Vikram Joshi	Bhopal	Tablet	Electronics	2026-08-29	3	22000	66000	Debit Card	Completed
ORD0100	Rahul Nair	Mumbai	Tablet	Electronics	2026-06-12	1	30000	30000	Cash	Cancelled